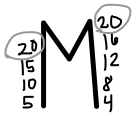
0

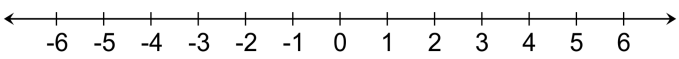
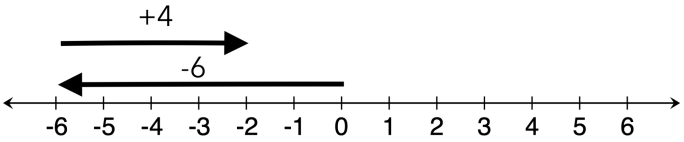
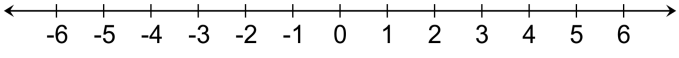
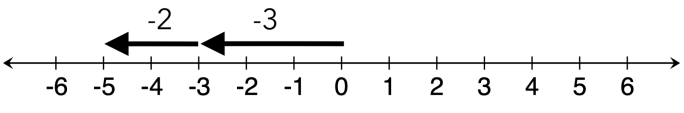
7 Math Unit 0 Fractions and Decimals STAR Cards

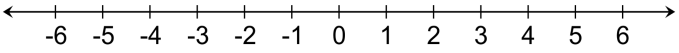
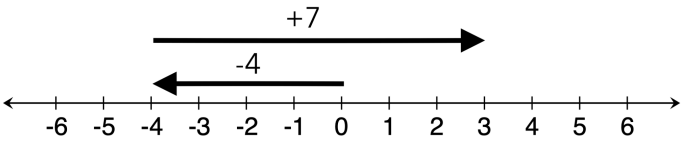
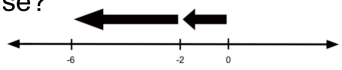
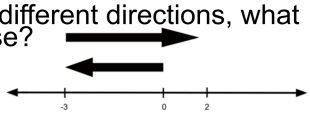
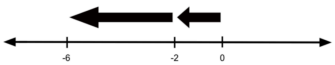
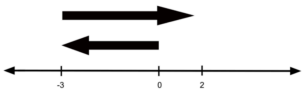
QUESTION	ANSWER
0a) ☆☆☆ What is a fraction?	0a) A fraction is a number that shows a **part of a whole**
0b) ☆☆☆ 1. What does the top number tell you? $\frac{3}{5}$ 2. What does the bottom number tell you?	0b) 1. How many **equal parts you have** (numerator) 2. How many **equal parts make up the whole** (denominator)
0c) ☆☆☆ Explain how to add (or subtract) fractions like $\frac{3}{5} + \frac{1}{5}$	0c) Because denominators are the SAME, add (or subtract) the numerators and keep the denominator. $\frac{3}{5} + \frac{1}{5} = \frac{4}{5}$

QUESTION	ANSWER
0d) ☆☆☆ Explain how to add (or subtract) fractions like $\frac{3}{5} + \frac{1}{4}$	0d) $\boxed{\frac{3}{5} + \frac{1}{4}}$ *Because denominators are DIFFERENT,* 1st **Find the LCM**  2nd **Make equal fractions** with the LCM as the denominators $\frac{3}{5} = \frac{12}{20}$ + $\frac{1}{4} = \frac{5}{20}$ 3rd **Add numerators, keep denominators** $\frac{12}{20} + \frac{5}{20} = \frac{17}{20}$
0e) ☆☆☆ Explain the rules for multiplying fractions with this example: $\frac{2}{3} \times \frac{4}{5}$	0e) **Multiply straight across** $\frac{2}{3} \times \frac{4}{5} = \frac{8}{15}$ simplify, if needed
0f) ☆☆☆ 1) How do we <u>simplify fractions</u>? 2) How can you simplify $\frac{6}{15}$?	0f) 1. To simplify fractions, **divide the top and the bottom by the same number.** 2. $\frac{6}{15} \frac{\div 3}{\div 3} = \frac{2}{5}$

QUESTION	ANSWER
0g) ☆☆☆ How do we convert a mixed number into an improper fraction? **Example:** $4\frac{1}{3}$	0g) 2nd) Add the numerator + 1 $4\frac{1}{3} = \frac{13}{3}$ x 3 1st) Multiply the whole number by the denominator (=12) whole number by denominator = 12 3rd) Keep the same denominator
0h) ☆☆☆ How do we change an improper fraction into a mixed number? **Example:** $\frac{10}{7}$	0h) Divide the numerator (top) by the denominator (bottom). Write your remainder as a fraction. $7\overline{)10} = 1\frac{3}{7}$ -7 3
0i) How do we <u>divide fractions</u>? Example: $\frac{1}{4} \div \frac{3}{8}$	0i) To divide fractions $\frac{1}{4} \div \frac{3}{8}$ **K**eep **C**hange **F**lip $\frac{1}{4} \times \frac{8}{3}$ *then multiply straight across* $\frac{1}{4} \times \frac{8}{3} = \frac{8}{12}$ *or* $\frac{2}{3}$

7 Math Unit 1 Integer Operations STAR Cards

QUESTION	ANSWER
1a) ☆☆☆ Before solving, how would you simplify these problems? 1. $5 + -3$ 2. $-7 - -2$	1a) 1. $5 + -3$ simplifies to $5 - 3$ ↩ 2. $-7 - -2$ simplifies to $-7 + 2$ ↩
1b) ☆☆☆ **Use a number line** to model this problem: $-6 + 4$ 	1b) $-6 + 4$  *The answer is -2*
1c) ☆☆☆ Use a number line to model this problem: $-3 + -2$ 	1c) $-3 + -2$  *The answer is -5*

QUESTION	ANSWER
1d) Use a number line to model this problem: $-4 - -7$  	1d) $-4 - -7$ \  *The answer is 3*
1e) Following the rules for multiplying and dividing with negatives, are these answers **positive** or **negative**? 1. $-\frac{1}{2} \times \frac{3}{4} =$ 2. $-a \times -b =$ 3. $4.8 \div -1.2 =$ 4. $323 \div 2 =$	1e) **Multiply and divide normally.** If signs match, positive. If signs don't match, negative. 1. $-\frac{1}{2} \times \frac{3}{4} =$ **NEGATIVE** 2. $-a \times -b =$ **POSITIVE** 3. $4.8 \div -1.2 =$ **NEGATIVE** 4. $323 \div 2 =$ **POSITIVE**
1f) When arrows point the same way, what operation do you use?  When arrows point different directions, what operation do you use?  	1f) Same Direction: ADD lengths  Different Directions: SUBTRACT lengths 

7 Math Unit 2 Expressions STAR Cards

2

QUESTION	ANSWER
2a) ☆☆☆ 1. What do the absolute value lines $\lvert\ \rvert$ mean? 2. How are the following problems different? $\lvert -2+4 \rvert$ vs $\lvert -2 \rvert + 4$	2a) 1. Distance from zero, ALWAYS POSITIVE 2. For $\lvert -2+4 \rvert$ **:Solve the inside**, then take the absolute value. For $\lvert -2 \rvert + 4$ **:Find the absolute value of -2**, then solve.
2b) ☆☆☆ How can you simplify an expression like this one: $4x + 3 + 2x + 5$	2b) **Combine like terms** Same variables go together, Normal numbers *go* together $4x + 3 + 2x + 5$ ↓ $6x + 8$

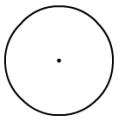
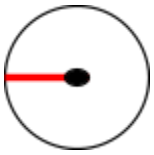
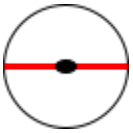
7 Math Unit 3 Equations STAR Cards

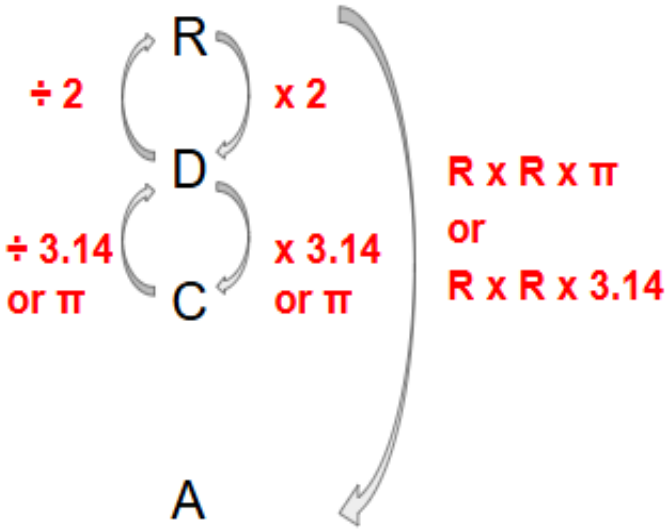
QUESTION	ANSWER
3a) ☆☆☆ What does it mean to <u>solve</u> an equation?	3a) *Solving* an equation means finding the **value of the variable that will make the left side equal the right side**.
3b) ☆☆☆ List the inverse operations for each: **I see** | **To undo it** | **I do** Addition | Subtraction | Multiplication | Division |	3b) **I see** | **To undo it** | **I do** Addition | Subtraction Subtraction | Addition Multiplication | Division Division | Multiplication
3c) ☆☆☆ How do you solve a 2 step equation like this? $6x + 3 = 27$	3c) 1st) Subtract 3 from both sides 2nd) Divide both sides by 6 $6x + 3 = 27$ $-3 \quad -3$ $\frac{6x}{6} = \frac{24}{6}$ $x = 4$

QUESTION	ANSWER
3d) How do you read each of these inequality symbols? 1. < 2. >> 3. ≤ 4. ≥	3d) 1. Less than < 2. Greater than > 3. Less than or equal to ≤ 4. Greater than or equal to ≥
3e) What is the difference between an inequality and an equation? $4x + 7 \geq 20$	3e) Inequalities have lots of answers; equations only have one answer.
3f) Write an inequality for each number line below: 1.  2.  3.  	3f) 1. $x < 6$  2. $x \leq 6$  3. $x > 8$ 

7 Math Unit 4 Circles STAR Cards

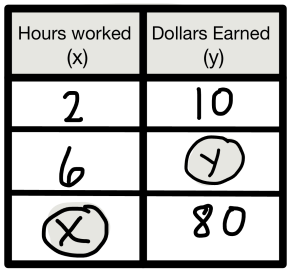
4

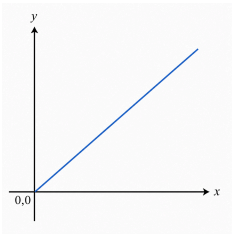
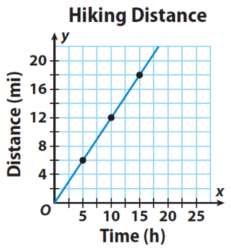
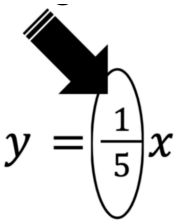
QUESTION	ANSWER
4A)  Define or draw the following parts of a circle: 1. Radius:  2. Diameter:  3. Circumference:  4. Area: 	4A) 1. Radius: **center** to the outside  2. Diameter: Side to side **through the center**.  3. Circumference: the **outside** “perimeter” of a circle.  4. Area: the **inside** of a circle. 

QUESTION	ANSWER
4b) Fill in the circle tool: R D C A	4b) 
4c) What are two ways you can write your answer with **pi**?	4c) 1. By using the symbol π *ex)* 10π 2. Estimated with **3.14** *ex)* 10 x 3. 14 = 31. 4
4d) Explain how you round a number to the hundredths place **8.562**	4d) 1st: Find the hundredths place *2nd:* Look one digit to the right *3rd:* If that digits **5 or more up the score,** 4 or less let it rest ↓ 8.562 Rounded = 8.56

5

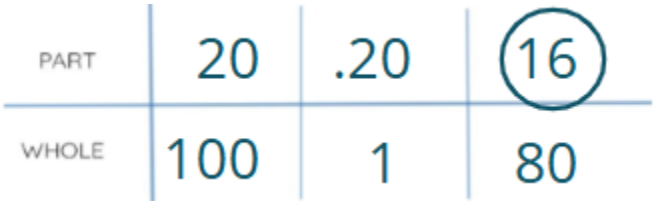
7 Math Unit 5 Proportional Relationships STAR Cards

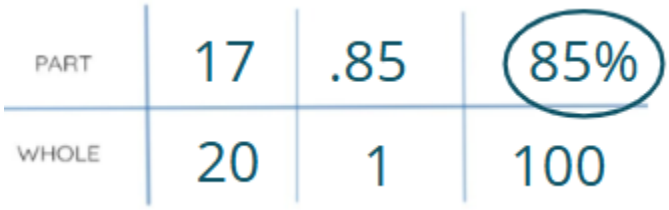
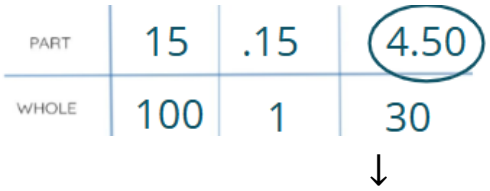
QUESTION	ANSWER
5a) ☆☆☆ What is a proportional relationship?	5a) A relationship where you are constantly **multiplying by the same number.**
5b) The table below is proportional. 1. How do you find the **C**onstant **O**f **P**roportionality in a table? 2. How do you find the missing **y** value in this Table? 3. How do you find the missing **x** value in this table?  	5b) 1. To find the C.O.P, divide $\frac{y}{x}$ 2. 6 x C.O.P = y 3. 80 ÷ C.O.P = x

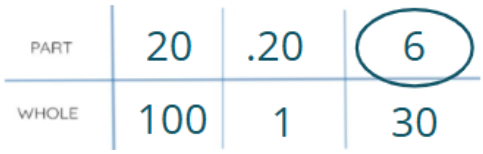
QUESTION	ANSWER
5c) ☆☆☆ 1. Sketch or describe what a proportional **graph** looks like 2. Describe how to find the **C.O.P** from a proportional graph	5c) 1. A straight line that goes through the origin (0,0)   2. Pick a point on the line and divide y by x (hint: when x=1, this is easiest!)
5d) ☆☆☆ Where can you find the **C**onstant **of** **P**roportionality in an equation? $y = \frac{1}{5}x$	5d) The C.O.P is the **number in front of the variable** 
5e) ☆☆☆ What are 3 main ways to represent a proportional relationship?	5e) Graph, Equation, Table

7 Math Unit 6 Percents STAR Cards

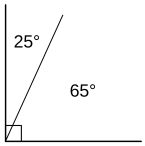
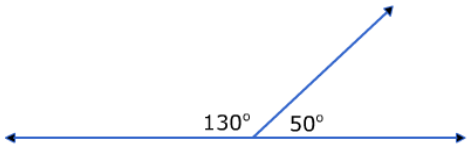
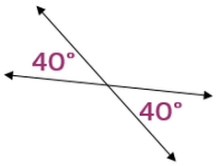
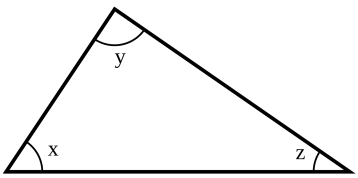
6

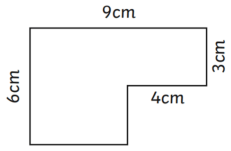
QUESTION	ANSWER
6a) ☆☆☆ 1. What is the definition of percent? 2. How do you write 34% as a fraction?	6A) 1. A percent is a fraction out of 100 2. 34% is $\frac{34}{100}$
6b) ☆☆☆ How can you find the **percent of** a number? Ex: Find 20% of $80	6b) Change the percent into a decimal and multiply it by the number. Ex: 

QUESTION	ANSWER
6c) ☆☆☆ If you need to find a percent in a real life situation how do you do it? *<u>Example:</u> Carlos got 17 out of 20 questions right on his science test.* ***What percent*** *did he get right?*	6c) **1st) Find the fraction** **2nd) Part divided by whole** **3rd) Decimal times 100** 
6d) ☆☆☆ Below is a problem where you need to find a **percent increase**. Explain how to solve it: *You go to a restaurant and your food costs $30. You want to add a 15% tip. What is the total amount you will pay?*	6d) <u>To calculate % increase:</u> 1st: Find the **percent of the number**  ↓ 2nd Add **whole** + % of the number $30 + $4.50 = **$34.50**

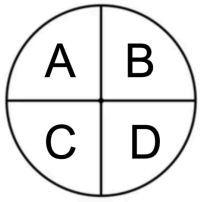
QUESTION	ANSWER
6e) ☆☆☆ How do you find the total after a % decrease? *Example: 20% discount on $30*	6e) <u>To calculate % decrease:</u> 1st) Find the **percent of the number** ↓  2nd) Subtract **whole** – % of the number $30 - $6 **= $24**
6f) ☆☆☆ How do you convert a …. Fraction to Decimal: Decimal to Percent: Percent to Fraction:	6f) How do you convert a …. Fraction to Decimal: **N**umerator ÷ **D**emoniator Decimal to Percent: Decimal ✖ 100 Percent to Fraction: Percent over 100

7 Math Unit 7 Geometry STAR Cards

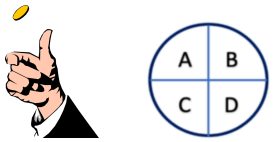
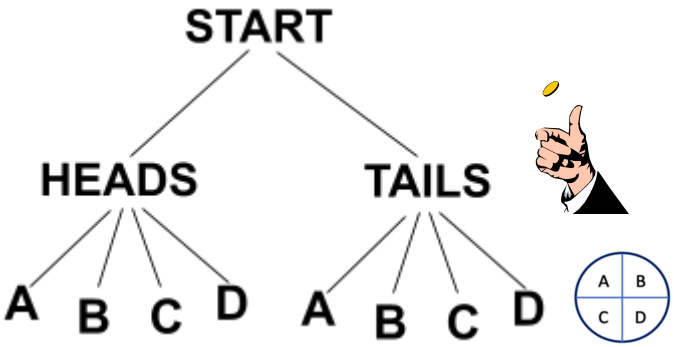
QUESTION	ANSWER
7a) ☆☆☆ 1.) Describe and sketch a complementary angle 2.) Describe and sketch a supplementary angle. 3.) Describe and sketch a vertical angle.	7a) 1. Complementary angles **add up to 90°**  2. Supplementary angles **add up to 180°**  3. Vertical angles are **opposite angles that are equal**. 
7b) ☆☆☆ What do the interior angles of any triangle in the world add up to?  	7b) The three angles in any triangle add up to 180. $x + y + z = 180°$

QUESTION	ANSWER
7c) ☆☆☆ How can you tell if three **lines** can form a triangle? Like these lines: _____ _____ _____	7c) The two smallest sides added together have to be bigger than the longest side. These lines would form a triangle! _____ _____ _____ __ + ___ > ___ S S L
7d) ☆☆☆ What is the difference between volume and surface area?	7d) Volume = The number of cubes that fit **inside** a **3D shape** Surface Area = The area of all the flat sides of **a 3D shape**
7e) How do you calculate the area of… • a rectangle? ▭ • a triangle? △ • a compound shape?  	7e) To find the area of a… ● <u>Rectangle:</u> **Length x Width** ● <u>Triangle:</u> **Base x Height ÷ 2** ● <u>Compound shapes:</u> 1.Break into shapes you know 2.Find the area of each 3.Add all areas together

7 Math Unit 8 Probability and Statistics STAR Cards

QUESTION	ANSWER
8a) ☆☆☆ 1. What is probability? 2. How do you find probability?	8a) 1. Probability is the **chance** something will happen 2. To find probability, write a fraction: $\frac{\text{Desired outcomes}}{\text{Total outcomes}}$
8b) ☆☆☆ If you spin this spinner 40 times how can you **predict how many times (probability)** you'll land on C? 	8b) Multiply the **Probability x total tries** $\frac{1}{4}$ x 40= 10 times
8c) ☆☆☆ 1. What is the difference between a <u>population</u> and a <u>sample?</u> 2. Why take a sample instead of surveying the whole population?	8c) 1. <u>POPULATION</u>: The **whole group** you are studying <u>SAMPLE</u>: a **small group** selected from the population 2. Surveying a sample is more efficient!
8d) ☆☆☆ What makes a <u>sample</u> **representative** and not **biased**?	8d) Representative samples: 1. Sample is **FROM the population** 2. Sample is as **BIG** as possible 3. Sample is **RANDOM** *Biased samples are not fair or favor one group*

QUESTION	ANSWER
8e) ☆☆☆ 1) What is a <u>measure of center</u>? Give 2 examples 2) What is a <u>measure of variability</u>? Give 2 examples	8e) 1) <u>Measures of center:</u> represent the **typical value** of the data set *Examples:* **mean, median, mode** 2) <u>Measure of variability:</u> measures how spread out your data is. *Examples:* **Range, MAD, IQR**
8f) ☆☆☆ How do you calculate Mean/Average?	8f) To calculate Mean/Average 1st) **Add** all the numbers 2nd) **Divide** by how many numbers you have
8g) ☆☆☆ How do you calculate Median?	8g) To calculate median: 1st) Put the numbers in **order** from least to greatest 2nd) Choose the **middle** number *If there is a tie between two numbers in the middle, add them and divide by 2*

QUESTION	ANSWER
8h) You win a game if you: ● Flip a coin to heads ● Spin this spinner and land on A  Sketch a tree diagram and write how many possible outcomes there are in this situation.	8h)  There are 8 possible outcomes